

AI Software Engineering Team Project Requirements

Project Overview

The AI Software Engineering Team platform helps software developers automate daily engineering tasks using artificial intelligence. Users can upload project documents, ask questions, generate code, and receive intelligent recommendations.

User Management

The system must support user registration and login functionality. Users should register using Name, Email and Password. Passwords must be securely hashed. The system should use JWT authentication for secure access to APIs.

Authentication Module

Authentication verifies user identity. Process: User enters email and password, password is verified, JWT token is generated, token is returned, protected APIs are accessed using Bearer token.

Database Module

The application uses PostgreSQL as the primary database. SQLAlchemy ORM is used for database operations.

Document Processing Module

Pipeline: File Upload, PDF Text Extraction, Text Chunking, Embedding Generation, Vector Storage.

Retrieval Augmented Generation (RAG)

RAG combines Vector Database, Semantic Search and Large Language Models. User question is converted to embeddings, relevant chunks are retrieved and sent to the LLM to generate answers.

Vector Database

The project uses ChromaDB. It stores chunks, embeddings and metadata and enables semantic search.

Cloud Deployment

The application can be deployed on AWS using EC2, S3, RDS and IAM.

Future Enhancements

AI Agents, Multi-Agent Collaboration, Code Generation, Bug Detection, Automated Testing, CI/CD Integration and Project Management Assistant.